Molecular Shapes (VSEPR)

Answer key with worked solutions

A. Understanding

1. VSEPR.

> Electron domains (bonds and lone pairs) around a central atom repel each other and arrange themselves **as far apart as possible**, which sets the shape of the molecule.

2. Domain geometry vs shape.

> The electron-domain geometry describes **all** domains, including lone pairs; the molecular shape describes only the positions of the **atoms**. Water has 4 domains (tetrahedral electron-domain geometry), but 2 are lone pairs, so its molecular shape is **bent**.

B. Predicting shapes

3. Shapes.

> (a) BCl_3: 3 domains, 0 lone pairs, **trigonal planar**, 120° (b) H_2S: 4 domains, 2 lone pairs, **bent**, less than 109.5° (c) PCl_3: 4 domains, 1 lone pair, **trigonal pyramidal**, less than 109.5° (d) CS_2: 2 domains (two C=S), 0 lone pairs, **linear**, 180° (e) SiH_4: 4 domains, 0 lone pairs, **tetrahedral**, 109.5°

4. Formaldehyde.

> Carbon has 3 domains (two C–H bonds and one C=O double bond, counted once) and no lone pairs: **trigonal planar**, H–C–H about **120°**.

5. Ammonium.

> N has 4 bonding domains and no lone pairs: **tetrahedral**, 109.5°.

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C. Bond angles

6. Trend.

> All three have 4 electron domains. Lone pairs repel **more strongly** than bonding pairs, so they push the bonds closer together. CH_4 has 0 lone pairs, NH_3 has 1 and H_2O has 2, so the angle decreases as the number of lone pairs increases.

D. Polarity

7. Polar or nonpolar.

> (a) CO_2: **nonpolar**; linear, so the two C=O polarities cancel (b) H_2O: **polar**; bent, so the O–H polarities don't cancel (c) CCl_4: **nonpolar**; tetrahedral with four identical bonds, so they cancel (d) NH_3: **polar**; trigonal pyramidal, so the N–H polarities don't cancel

E. Challenge

8. XeF_4.

> Six domains point to the corners of an octahedron. The two lone pairs go **opposite each other** (180° apart) to minimise lone pair–lone pair repulsion, leaving the four F atoms in a square: **square planar**, F–Xe–F 90°.

9. SF_4.

> Five domains with one lone pair, which sits in an equatorial position: **seesaw** shape.

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